Note

From Newton's 2nd law, $\vec{F} = m\vec{a}$ Here, the net force on the L.H.S is $T\text{-}F_{as}$

In R.H.S, $m\vec{a}$ is equivalent to the centripetal force = $\frac{mv^2}{l}$ which makes the bob oscillate.

From the Figure 10.21, we can observe that the tangential component W_{ps} of the gravitational force always points towards the equilibrium position i.e., the direction in which it always points opposite to the direction of displacement of the bob from the mean position. Hence, in this case, the tangential force is nothing but the restoring force. Applying Newton's second law along tangential direction, we have

$$m\frac{d^2s}{dt^2} + F_{ps} = 0 \Rightarrow m\frac{d^2s}{dt^2} = -F_{ps}$$

$$m\frac{d^2s}{dt^2} = -mg\sin\theta \qquad (10.51)$$

where, s is the position of bob which is measured along the arc. Expressing arc length in terms of angular displacement i.e.,

$$s = l\theta \qquad (10.52)$$

then its acceleration,

$$\frac{d^2s}{dt^2} = l\frac{d^2\theta}{dt^2} \qquad (10.53)$$

Substituting equation (10.53) in equation (10.51), we get

$$l\frac{d^2\theta}{dt^2} = -g\sin\theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\sin\theta \qquad (10.54)$$

Because of the presence of sin θ in the above differential equation, it is a non-linear differential equation (Here, homogeneous second order). Assume "the small oscillation approximation", sin $\theta \approx \theta$, the above differential equation becomes linear differential equation.

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\theta \qquad (10.55)$$

This is the well known oscillatory differential equation. Therefore, the angular frequency of this oscillator (natural frequency of this system) is

$$\omega^2 = \frac{g}{l} \qquad (10.56)$$

$$\Rightarrow \quad \omega = \sqrt{\frac{g}{l}} \text{ rad } s^{-1} \qquad (10.57)$$

The frequency of oscillations is

$$f = \frac{1}{2\pi}\sqrt{\frac{g}{l}} \text{ Hz} \qquad (10.58)$$

and time period of oscillations is

$$T = 2\pi\sqrt{\frac{l}{g}} \text{ second} \qquad (10.59)$$

Laws of simple pendulum

The time period of a simple pendulum

a. Depends on the following laws

(i) Law of length

For a given value of acceleration due to gravity, the time period of a simple pendulum is directly proportional to the square root of length of the pendulum.

$$T \propto \sqrt{l} \qquad (10.60)$$

(ii) Law of acceleration

For a fixed length, the time period of a simple pendulum is inversely proportional to square root of acceleration due to gravity.

$$T \propto \frac{1}{\sqrt{g}} \qquad (10.61)$$

b. Independent of the following factors

(i) Mass of the bob

The time period of oscillation is independent of mass of the simple

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